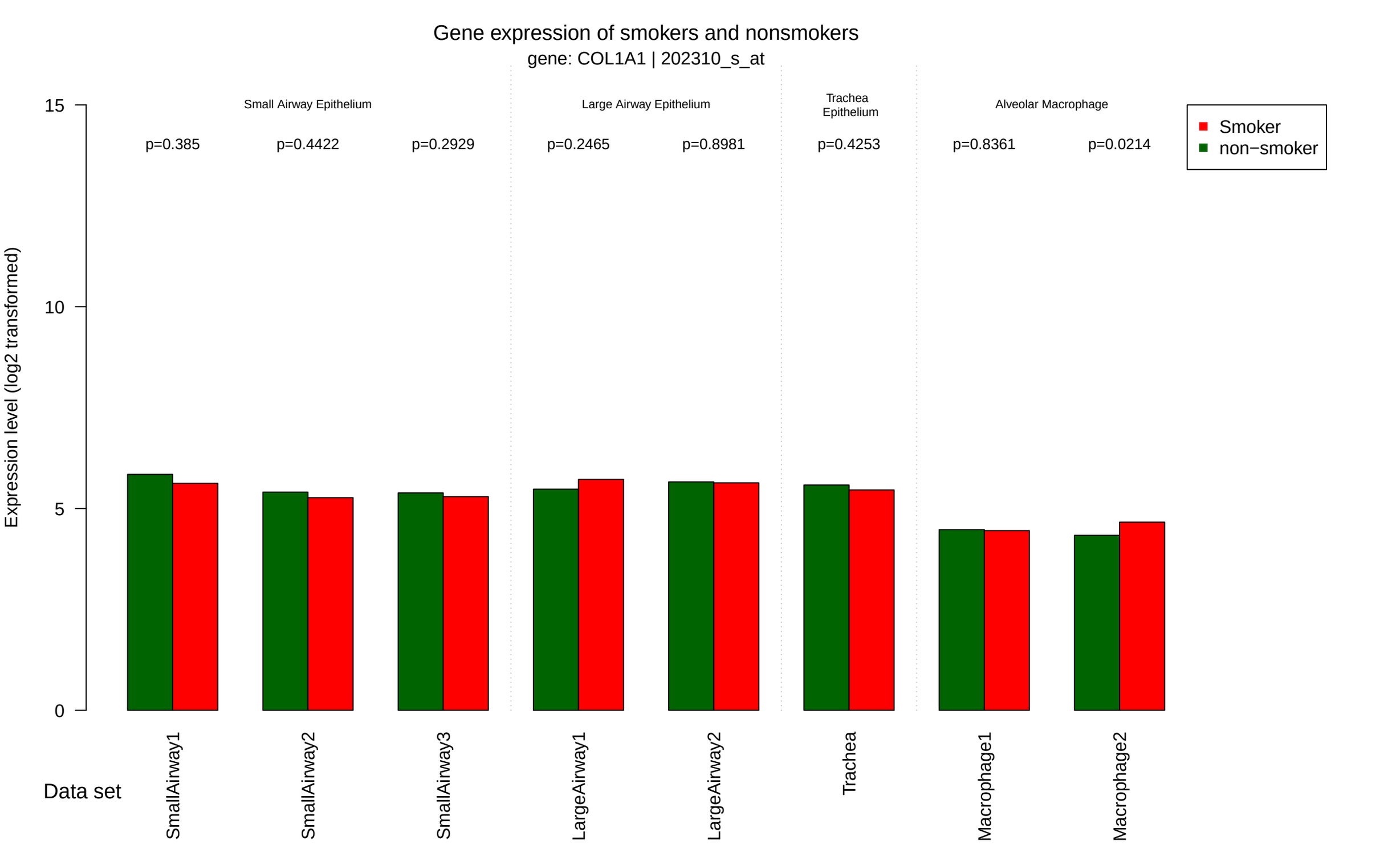
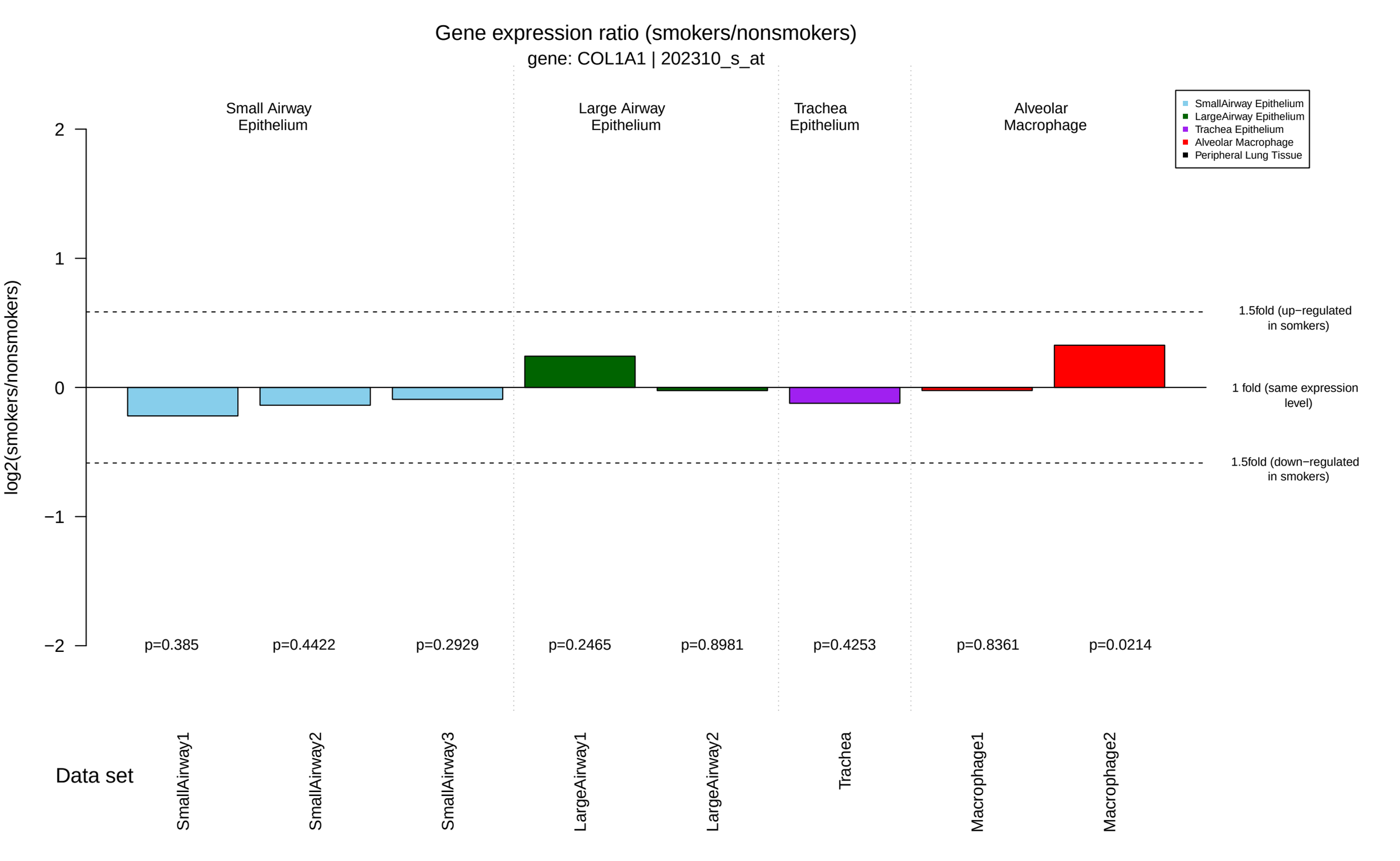
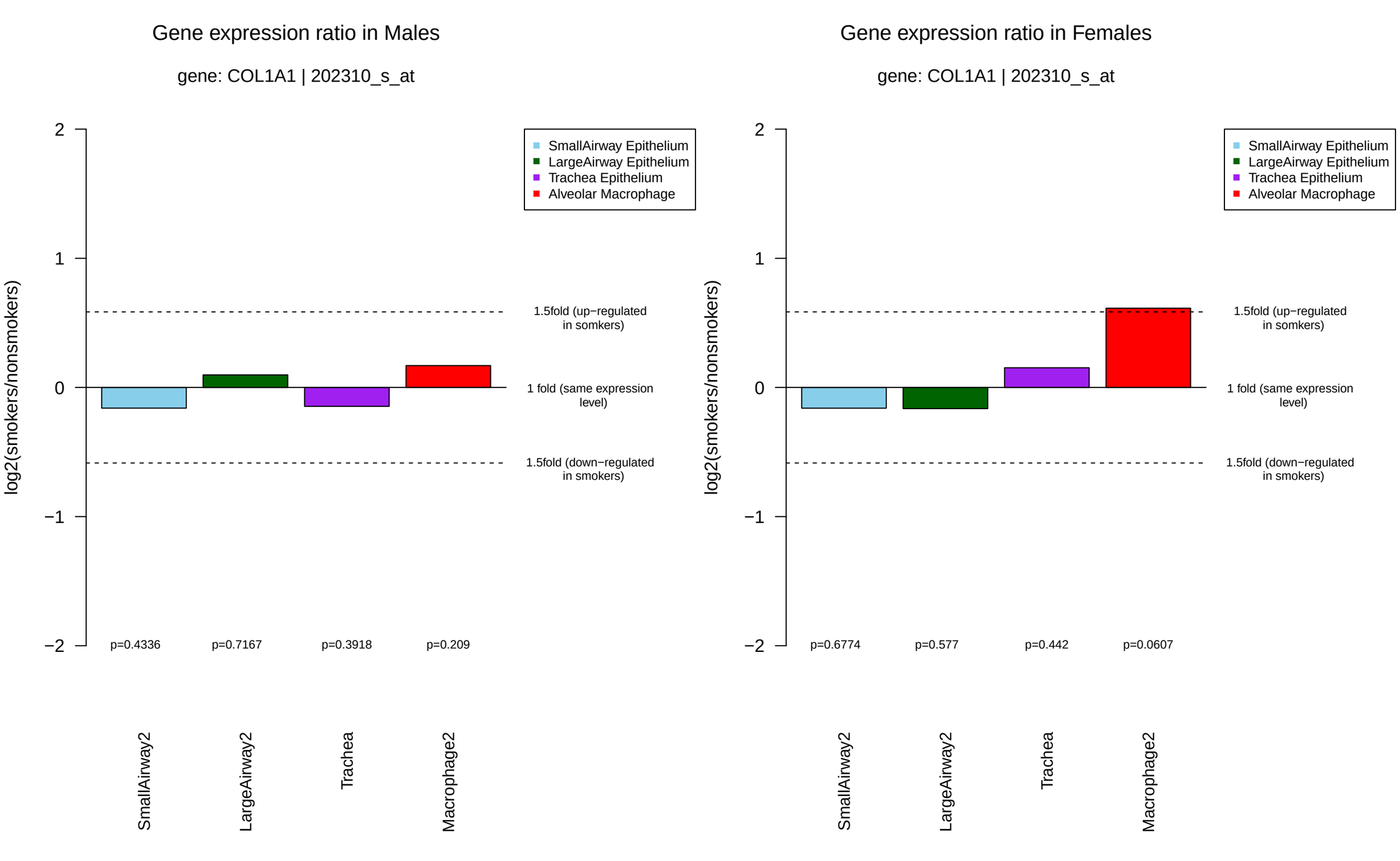
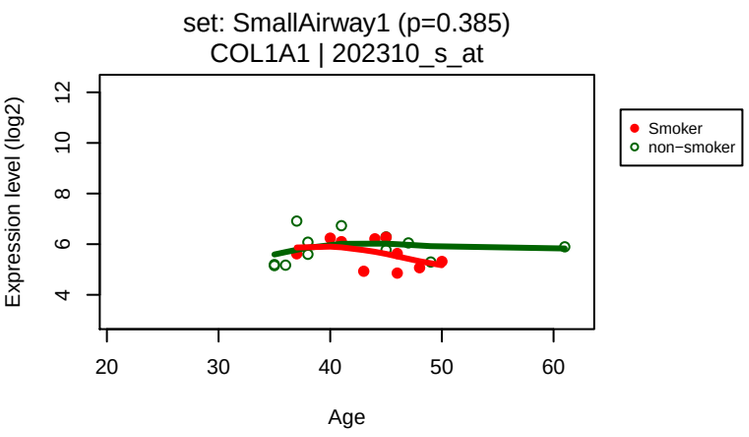
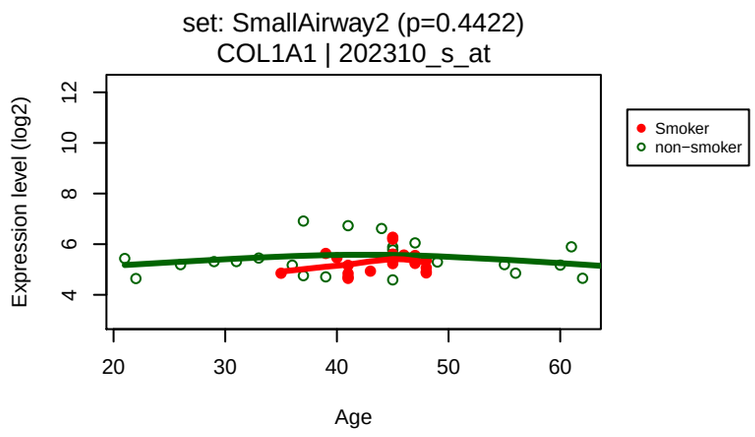
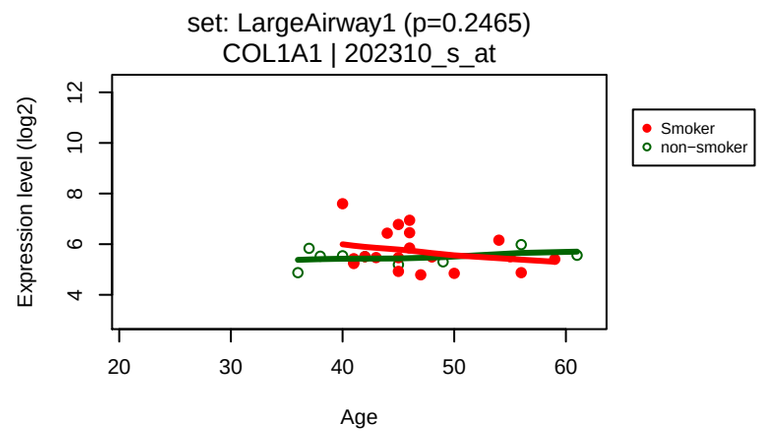

Gene expression of smokers and nonsmokers
gene: COL1A1 | 202310_s_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.385
p=0.4422
p=0.2929
p=0.2465
p=0.8981
p=0.4253
p=0.8361
p=0.0214
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

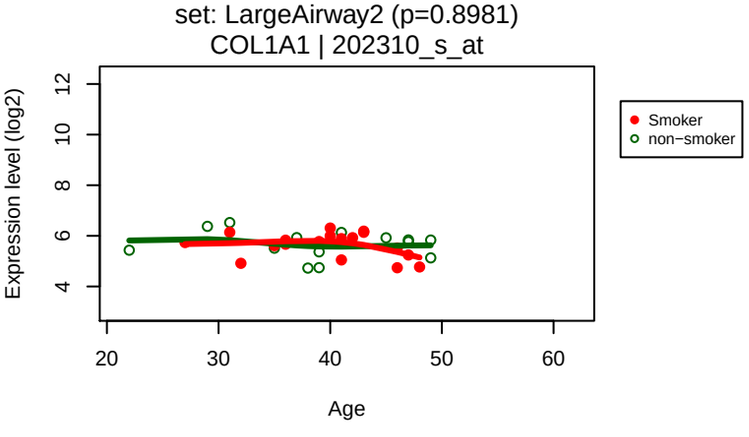

Gene expression ratio (smokers/nonsmokers)
gene: COL1A1 | 202310_s_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.385
p=0.4422
p=0.2929
p=0.2465
p=0.8981
p=0.4253
p=0.8361
p=0.0214
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

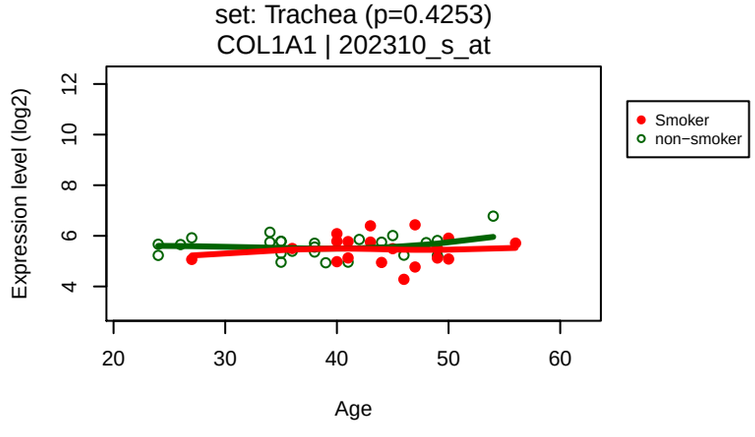

Gene expression ratio in Males
gene: COL1A1 | 202310_s_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.4336
p=0.7167
p=0.3918
p=0.209
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: COL1A1 | 202310_s_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.6774
p=0.577
p=0.442
p=0.0607
SmallAirway2
LargeAirway2
Trachea
Macrophage2

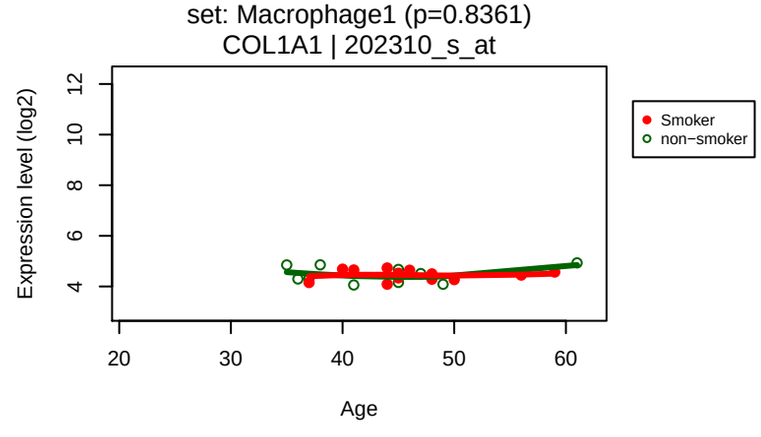

set: SmallAirway1 (p=0.385)
COL1A1 | 202310_s_at
Expression level (log2)
Age
Smoker
non−smoker

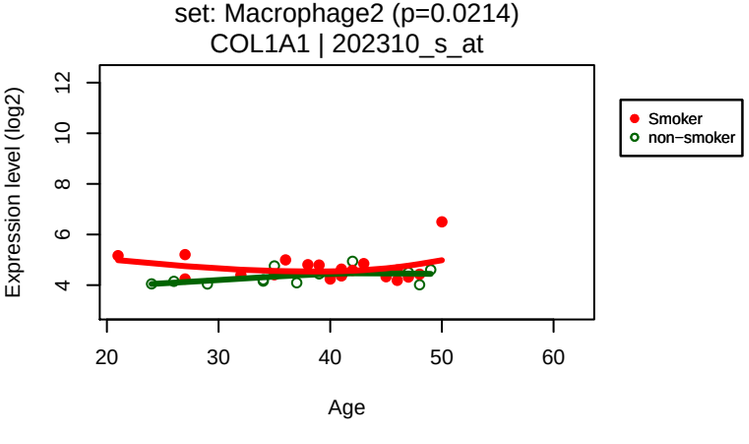

set: SmallAirway2 (p=0.4422)
COL1A1 | 202310_s_at
Expression level (log2)
Age
Smoker
non−smoker

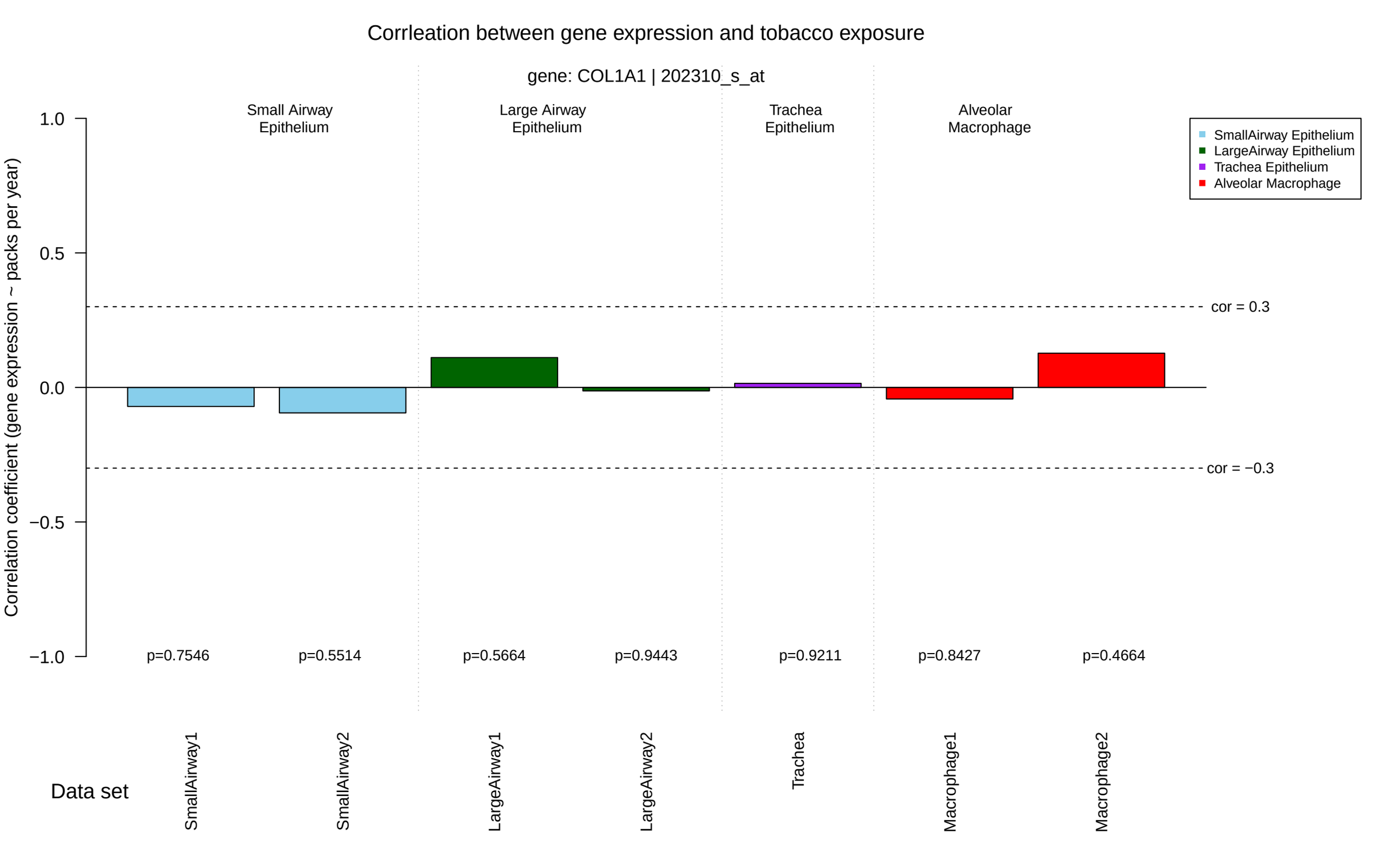

set: LargeAirway1 (p=0.2465)
COL1A1 | 202310_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.8981)
COL1A1 | 202310_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.4253)
COL1A1 | 202310_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.8361)
COL1A1 | 202310_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.0214)
COL1A1 | 202310_s_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: COL1A1 | 202310_s_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.7546
p=0.5514
p=0.5664
p=0.9443
p=0.9211
p=0.8427
p=0.4664
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2